

for programmable power supplies with features like constant-current source, constant-voltage source and frequency locking.

"3½- or 4½-digit handheld digital multimeters with basic functions of voltage, current and resistance measurement and component testing are required." Function generators up to 2 MHz and capable of generating basic sine, square, triangular and pulse waves are used in basic lab setups, whereas you need arbitrary waveform generators with higher bandwidths in advanced labs.

"The basic lab equipment need to be in the simplest and primitive form in order to facilitate learning," says Anand Bhushan, managing director, Bhushan & Bhushan. He adds, "Basic equipment should be manual in order to make the engineers understand the basic concepts. Automatising can be done to higher level of equipment."

Manufacturers are working towards making basic lab equipment more and more affordable, reliable and easily available. "We try to guide them towards the right balance between requirements and prices as many a times academia is unaware of availability and price of the equipment in the market," says Manish Kwatra, CEO, Metro Electronic Products. Maintenance and service support is another focus area of many T&M manufacturers. Kwatra

What do the T&M manufacturers offer?

"We try to guide the institutes towards the right balance between requirement and price"—*Manish Kwatra, CEO, Metro Electronic Products*

"We make sure that the equipment has proper grounding and proper cabinet coating to ensure safety"—*T.N. Singh, CEO, Salicon Nano Technology*

"The products are designed to impart knowledge and provide hands-on experimentation to every learner"—*Mohammed Ghouse, Manager, Scientech Technologies*

"As the lab equipment are used by many users for different experiments, these should have operational ease as per multiuser needs"—*Anirudh Kela, Managing Director, Scientific-Mes Technik*

"We bring very distinct advantage to labs by providing not only 'professional grade' test & measurement equipments but also complete set of experiments, practicals and teaching notes that meet every budget need"—*Arun Dogra, Vice President-Sales, Marketing and Service, Agilent Technologies*

"It is very important to impart training to the lab assistants, instructors and teachers and we welcome the institutes to get their staff trained"—*Anand Bhushan, Managing Director, Bhushan & Bhushan*

adds, "We provide robust equipment with in-built protections as many a times students are not aware of the right way of handling them."

Advanced technical labs

Colleges and institutes offering degree programmes in engineering have subject-specific labs. For example, for signals and systems, the experiments deal with high-frequency signals. To view such signals, you require digital oscilloscopes with high bandwidth. Also, you will need spectrum analysers, which you may not require in the power electronics lab.

Technical institutes and T&M manufacturers work closely in setting up these labs, providing training to the staff and also selecting list of experiments that will help the students to get comprehensive feel of the subject.

"The secondary set of equipment that facilitates experimentation includes mixed-signal oscilloscopes, logic analysers, spectrum or network analysers, frequency counters, programmable power supplies and arbitrary function generators," explains Anirudh Kela, managing director, Scientific-Mes Technik. "These are economical and customisable lab worksta-



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